

Unsupervised Technique for Anomaly Detection in Qatar Stock Market

Haya Al-Thani, Hanadi Hassen, Somaya Al-maadeed, Noora Fetais, Ali Jaoua
Computer Science and Engineering Department
College of Engineering, Qatar University
Doha, Qatar
200662244, hm1409611, s_alali, n.almarri, jaoua @qu.edu.qa

Abstract— The aim of anomaly detection is to find patterns or data points that are not confirming the expected behavior inside the dataset. Techniques from a variety of disciplines like machine learning, statistics, information theory and data mining are used to solve this problem. The form of input data from stock market is a non-linear complex time series. Hence, the statistical methods in this case will be ineffective. Using the behavior of similar time series for detecting anomalies in Qatar stock exchange and American stock market index (Standard & Poor (S&P)) is the main goal of this paper. Supervised learning techniques were used extensively in detecting stock market manipulation. The problem of supervised learning techniques is that they require substantial effort in labeling the data. Having dynamic nature of anomalous behavior causes another problem. In this research, we investigate the use of unsupervised learning for detecting stock market manipulation and we introduce a new preprocessing step for improving the recall of the anomaly detection system without hurting the precision. The Contextual Anomaly Detector (CAD) that is based on unsupervised technique is used to find anomalies by looking at time series that have similar behaviors. The use of our new preprocessing steps with CAD improved the recall significantly compared to other studies.

Keywords— Anomaly Detection; Supervised Learning; Unsupervised Learning.

I. INTRODUCTION

Anomaly detection has been widely researched because it has a variety of application domains such as intrusion detection, cyber-security, credit cards, fraud detection or healthcare. The aim of anomaly detection is to find patterns or data points that are not confirming the expected behavior inside the dataset. They are called anomalies or outliers.

Significant information can be reflected using anomaly detection, for example, the presence of a tumor can be discovered as an anomaly in an MRI scan. Furthermore, discovering anomalies in network traffic may indicate a hacking activity. This significant information contributes to the importance of anomaly detection. The diversity of applications in this area has led to intensive research in anomaly detection in statistics and computing. In this research we will use stock market data and identify market manipulations using anomaly detection.

The simplest way to detect anomalies is to define a range reflecting normal behavior in the data. Any data point that is outside this range will be classified as an anomaly. Although

this approach is simple, it can be very challenging because defining the range for every instance representing normal behavior is difficult. Discovering abnormal behaviors that are close to the boundary of normal behaviors is hard. In addition, malicious actions usually tend to reproduce the normal behaviors, hence, detecting anomalies of this type is very hard. The changing nature of normal behavior represents another challenge in that the current normal behavior may change to abnormal behavior in the future.

For different applications, applying one technique for defining anomalies is difficult, because in some applications, small deviation may be very significant like in healthcare while in the stock market these small deviations may not be significant and can be treated as normal. Another challenge in anomaly detection is the absence of the labeled training data in addition to the noise in the data that is undistinguishable from the actual anomalies.

These challenges were researched from diverse disciplines such as statistics, data mining, information theory and spectral theories. The input data itself is the major factor for selecting the suitable anomaly detection technique. The way in which the instances of the data are related to each other, as well as the nature of the data itself, serve as a cue for the necessary action to be taken.

There are many types of data instances: they can be sequential, where instances are ordered, like; time-series data, protein sequences or genome data, spatial data, where data instances are in relation with their neighbors' data, like; geographical data, traffic data or geological data or graphical data, where the data is represented as vertices. Before defining anomaly, understanding the nature of the data is very important.

In the following sections we will explore the literature for anomaly detection, our proposed system and the results we have achieved.

II. LITERATURE REVIEW

Anomaly detection using supervised methods requires labeled data which can be used in building the predictive model for classifying normal and abnormal activities. After building the model, it is used to determine the class of the new unlabeled data. Acquiring representative and accurate labeled data is a challenging task. Compared to the normal data, the

anomalous data tend to be fewer which causes an imbalance in the training datasets.

H. Öğüt, M. Mete Doğanay, and R. Aktaş [1] applied a supervised learning technique for detecting market manipulation. Using Support Vector Machines (SVMs) and Artificial Neural Networks (ANNs), they build their model which outperforms the statistical techniques. D. Diaz, B. Theodoulidis, and P. Sampaio [2] applied tree generating learning methods, QUEST, C5.0 and C&RT to detect intraday price manipulation after they constructed a labeled dataset from US Securities and Exchange Commission during 2009.

K. Golmohammadi, O. R. Zaiane, and D. Díaz [3] extended the work in [2] using the same dataset and applied different classification methods for detecting market manipulation. In their experiments, the Naïve Bayes classifier achieved F2 measure of 53%, surpassing the results achieved by using other classifiers.

The unsupervised anomaly detection is more applicable than the supervised method because it does not need a labeled dataset. Moreover, it assumes that the volume of normal data points is bigger than the volume of anomalous data points. Some examples of unsupervised techniques applied in fraud detection include regressing, rule induction, Hidden Markov Model (HMM), visualization and anomaly detection.

Linear regression, autoregressive integrated moving average (ARIMA), and auto-regression moving average (ARMA) are used in detecting manipulations and forecasting stock market trends [4][5][6]. These statistical models are not able to handle non-linear, irregular and noisy data.

Logistic regression is applied to detect market manipulation by [7] and it showed a better prediction rate than linear regression model.

Rule induction is one of the data mining methods that were applied for detecting fraud in the stock market. It is very popular among security market investigators and auditors. K. Jungwon, A. Ong, and R. E. Overill [8] implemented a rule induction method for fraud detection. The rules were generated randomly using association rules algorithm apriori. These rules were applied to a database containing legitimate transactions and they disregarded any rules matching the data. The remaining rules are used for monitoring new data. Then the rules that detect anomaly are replicated by introducing small random mutation. For anomaly detection, they use all successful rules.

Another use of association rules is presented in [9]. They are applied for detecting manipulations intraday trades in the Thai bond market.

The Hidden Markov Model (HMM) is used in anomaly detection and market manipulation. Anomaly detection in intrusion detection systems (IDS) is investigated by [10]. The method they applied assumes that the normal time series are generated by the hidden Markov process. Using the market data of US and UK, [11] track the changes of the time series using Adaptive Hidden Markov Model with Anomaly States (AHMMAS).

Considering the sellers, buyers and the relationship between them in the stock market data is interesting. The social network generated by this relationship is presented by nodes and the edges where the nodes are the entities and edges are the relationships. Using this type of networks, [12] identified fraudulent accounts in an exchange using interactive visualization and Social Network Analysis (SNA).

Techniques used for detecting an anomaly search for data that is dissimilar to the majority of the data observations. This idea is applied by [13]. They used Peer Group Analysis (PGA) to find outliers in Bangladesh stock market data. The PGA categorizes similar objects into peer groups. Then the centroid of each group is calculated and the distance of group members from their group centroid is also calculated. Group members with the largest distance from their group centroid are classified as outliers.

K. Golmohammadi and O. Zaiane [14] introduced an unsupervised method for market manipulation. They extended the work in [15] and developed Contextual Anomaly Detector (CAD) for prediction. Their solution assumes that stocks behave and react similarly if they are within the same sector. This method improved recall and outperformed k-nearest neighbor and random walk.

III. SYSTEM OVERVIEW

Using similar time series, [14] identified market manipulations. Our aim in this paper is to improve their work. In their paper [14], they proposed Contextual Anomaly Detector (CAD) to predict anomalies. Firstly, the time series inside the same industry sector were selected. Then, the centroid for each subset is calculated by averaging time series. They calculated the correlation between the centroid of the subset and the time series to find the predicted value for that time series. The anomaly is discovered for a point if there is a deviation with certain threshold between the actual value and the predicted value.

In this paper, we added a new preprocessing step to the proposed CAD solution to improve the previous work's recall without losing precision. We propose to use the Simple Moving Averages (SMA) of the companies' price changes.

Our proposed system is illustrated in Fig. 1. Starting with clean stock market data free of manipulation, our proposed method will be tested. Then, we will insert anomalies in the clean data to evaluate the system. At this point, we will apply our proposed preprocessing step using SMA. Anomalies will

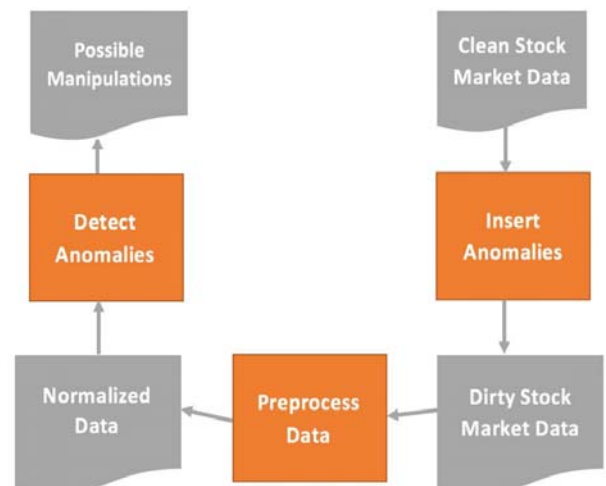


Fig. 1. General overview of the system.

be detected by the centroid and prediction method. Possible manipulation instances will be returned by the system.

IV. METHODS

Market manipulation or anomaly could be present if geocompanylar companies, a company behave differently than the group. To prove the similarity between companies of the same sector, we first need to understand the data. Thereafter, we can apply our preprocessing and identify anomalies. For evaluation, we will insert anomalies.

A. Data Collection

Two datasets were tested, the first dataset is used by [14] and the second dataset is collected from Qatar Stock Market (QSM). The time aspect is the first attribute in both datasets. The names of the companies and the closing price form the other attributes.

The first dataset is divided into industry sectors of the American stock market index S&P 500 which consists of the largest 500 companies. This index is representing US stock market and it is one of the most followed indices. It is monitored by regulatory organizations for that it is assumed to be manipulation free. The second dataset which belongs to Qatari companies is extracted using Bloomberg API and it is divided the same way as S&P 500 dataset.

B. Correlation Study

The correlation study is conducted to investigate whether companies within the same sector have similar behaviors or not. Correlation coefficients are calculated to test if two variables move the same way. Equation (1) shows how these coefficients are calculated using Pearson product-moment correlation ρ_{xy} for the two series r_x and r_y . Where the cov is the covariance and σ_x is the standard deviation.

$$\rho_{xy} = \frac{cov(r_x, r_y)}{\sigma_x \sigma_y} \quad (1)$$

In financial applications, this statistic can be used to determine how two funds are related to each other. It can also help investors diversify their gains by adding negatively correlated fund to their portfolio. This measure is applied to establish whether the sectors' time series behave similarly.

An example of correlation matrix is shown in TABLE I. using S&P consumer staples dataset. The coefficient of the correlation between companies is close to one and this indicates the similarity between companies. To monitor the possible manipulation, this proportionality can be used.

The S&P index is comprised of settled organizations. Most sectors are comprised of exceedingly correlated companies. The Information Technology sector does not follow this supposition. TABLE. II contains examples of the IT segments correlations. Companies of this sector are varied such as Cisco, Apple and Visa. According to the variation between these companies, the correlation between them is less.

Well-established companies and highly correlated are listed in the S&P exchange. However, for newer and younger markets like the Qatar stock market, this correlation does not hold. A minor correlation between QSM sectors is found based on similar studies. The correlation matrix shown in TABLE. III is for QSM companies in consumer goods and services sector. Due to the immaturity and other differences between these younger companies, their trends differ.

C. Preprocessing

Before detecting anomalies, a normalization step for time series data is required. When monitoring the market for manipulation, the most important feature is the stock closing prices which has a wide range. Relating the actual stock price to other companies is not of interest. It is important to determine how a particular company's price changes relative to other companies. Even if prices of similar companies are varied, they experience similar price changes.

Equation (2) illustrates price percentage change Ch_t where t is the time and P is the price. This preprocessing step is proposed in [16] for scaling stock prices and normalizing the data.

$$Ch_t = \frac{(P_t - P_{t-1})}{P_{t-1}} \quad (2)$$

There is a dependency between each row and the one before. Information that reflects the history of the stock beyond that is not provided. To improve results, we propose to change the way the data is preprocessed. while stock price movement's history is still reflected, the data can be normalized.

We propose to use a Simple Moving Average (SMA) using (3) where t is the time, Ch is the stock price change, Ch_0 is the initial price change and n is the number of instances in the dataset.

$$SMA_t = \frac{(Ch_t + Ch_{t-1} + Ch_{t-2} + \dots + Ch_0)}{n} \quad (3)$$

In addition to data scaling and normalization, this formula maintains stocks' history information. We assume this preprocessing step will improve recall greatly.

TABLE I. THE CORRELATION MATRIX FOR S&P CONSUMER STAPLES

Company	Walmart	Walgreen	Coca Cola	Colgate-Palm	General Mills
Walmart	-	0.95	0.89	0.96	0.94
Walgreen	0.95	-	0.86	0.93	0.91
Coca Cola	0.89	0.86	-	0.90	0.90
Colgate-Palm	0.96	0.93	0.90	-	0.98
General Mills	0.94	0.91	0.90	0.98	-

TABLE II. CORRELATION MATRIX FOR S&P IT SECTOR

Company	FACEBOOK CLASS A	GOOGLE 'A'	CISCO SYSTEMS	APPLE	INTEL	VISA 'A'
FACEBOOK CLASS A		0.86	0.42	-0.05	0.41	0.61
GOOGLE 'A'	0.86		0.28	0.80	0.35	0.96
CISCO SYSTEMS	0.42	0.28		0.20	0.92	0.26
APPLE	-0.05	0.80	0.20		0.32	0.76
INTEL	0.41	0.35	0.92	0.32		0.61
VISA 'A'	0.84	0.96	0.26	0.76	0.61	

D. Anomaly Detection

To detect anomaly, a new sample is compared with a set of normal samples. Then, those two are measured against one another and the new sample is assigned an anomaly score. Samples with high scores will be classified as anomalies. Distinguishing between the normal and abnormal behaviors in a complicated field like finance is not straightforward.

Fraud detection in stock market using supervised learning provided good results. However, it depends on the existence of the labeled data which is impractical. The labeling process might need to go through litigation cases and use these observations as anomalies and the rest of the data as normal. Another way is to synthesize the data which will not reflect the manipulations in real life.

The unsupervised anomaly detection is used to overcome the problem of manual data labeling. Contextual Anomaly Detection (CAD) is proposed by [14] to detect manipulation through the prediction of series next value using the behavior of similar series.

In the proposed CAD method by [14] a subset of time series is taken based on a window size and a centroid is calculated for this subset by taking the mean at every time instance. The expected behavior of this subset is represented by this centroid, which will be used with the feature of each time series, to predict the value of that time series. Fig. 2 shows how the centroid appears in a given company time series set.

Equation (4) illustrates how the predicted value of a time series X_t at time t is calculated using Pearson correlation ($\text{cor}(X_t, C)$) where C is the centroid.

$$\hat{x}_{tt} = x_{tt-1} * \text{cor}(X_t, C) \quad (4)$$

E. Anomaly Insertion

To evaluate our system, we will insert synthesized anomalies in the datasets. The S&P dataset is manipulation free and has been tested in many research studies because it is ideal for testing fraud detection or forecasting [1][14] [17] [18].

Manipulations are represented as outliers or anomalies in a collection of normal stock behavior. It can appear as an abnormal drop or jump in the price of the company. By defining the outliers, the manipulation free data will be injected with the manipulation cases for evaluation. To compare our work with others, we applied the same injection method in [14]. This method used for time series does not follow normal distribution and it is adopted from Turkey's

method [19]. The outlier instance in a normally distributed time series is defined as three standard deviations away from the mean.

In other distributions with skewness, the outlier $O(x_{it})$ for the mean, Q_1 , Q_3 are the 25th percentile (1st quartile) and 75th percentile (3rd quartile) respectively, IQR is the inter-quartile range and σ is the standard deviation.

$$O(x_{it}) = \begin{cases} \mu + (Q_3 + (3 * IQR)) \text{ if } \gamma > \epsilon, \text{ upperbound} \\ \mu - (Q_1 - (3 * IQR)) \text{ if } \gamma > \epsilon, \text{ lowerbound} \\ \mu \pm 3\sigma \text{ if } \gamma = 0, \text{ if normal distribution} \end{cases} \quad (5)$$

To find outliers for series with skewed distribution, the statistical center of the range is calculated. This is done by using the first and the third quartiles. Equation (6) is used to find the skewness of the series where μ_2 and μ_3 are the second

and third central moments about the mean.

$$\gamma = \frac{\mu_3}{\mu_2^{3/2}} \quad (1)$$

Outliers are defined independently; each time series is independent from other series. The synthetic outliers are not affecting the detection method because the training data is used for the learning process. It is used for evaluating the SYSTEM.

V. EXPERIMENTAL RESULTS

The new preprocessing step is tested, and the results in TABLE. IV shows how the recall improved significantly using the SMA preprocessing step. The table also illustrates the results of the application of k-means clustering and the application of CAD as in the original paper [14] and when combined with SMA.

The below performance in our experiments is gained by using 0.1% anomalies and a window size of 15. For different time instances these measures are also stable.

In problems like market manipulations, the recall is very important. The preprocessing step proposed here improved recall and precision compared to other two algorithms. Although the precision is low, almost all anomalies were

TABLE III. CORRELATION MATRIX FOR CONSUMER GOODS AND SERVICES SECTOR IN QSM

Company	ZAD Holding	Q.German Medical	Qatar Cinema	Medicare	Salam International
ZAD Holding		0.58	-0.07	0.93	0.59
Q.German Medical	0.58		0.22	0.58	0.57
Qatar Cinema	-0.07	0.22		0.04	0.44
Medicare	0.93	0.58	0.44		0.60
Salam International	0.59	0.57	0.44	0.60	

retrieved. The cost of false positives and false negatives is not equal. It is undesirable to get false positives, but it can be very expensive for companies to miss manipulations.

The value at $t-1$ using SMA preprocessing carries information of historical movement of the time series, so that the prediction of the time series at t is more accurate. However, the precision using this method is low because the anomalous value propagates to the following series data. The changes are averaged over time, for that, the anomaly is carried into subsequent entries resulting in normal entries labelled as anomalies. This problem is not present in the original preprocessing method because anomalous value affects only the value immediately after it.

To overcome the problem of our method, a range of certain number of rows would be involved in SMA calculation. For the value of the time series at t , the SMA is calculated from $t-10$ to t . This will propagate the anomaly only to the next ten rows. Another method to weaken anomaly effects over time would be the exponential moving average where the latest data is assigned with more weights.

Almost in all sectors, better precision and recall is achieved by k-means than CAD. In our experiments the IT sector does not perform well because as we explained earlier in the correlation study the companies in this sector are less correlated (see TABLE. II). Data clustering in this sector did not achieve same results as in other sectors. If the data could be spherically clustered, the k-means clustering achieves better results. K-means is centroid based clustering and it performs better than density based clustering. However, for non-global data like financial data, centroid based clustering will not work well as we showed in the IT sector [6].

A. Varying Anomaly Percentage

TABLE. IV shows our results after inserting 0.1% anomalies and TABLE. V shows the results of CAD-SMA using an

increasing number of anomalies.

There is an inverse relationship between the number of anomalies and the recall. The recall decreases with the increase in the number of anomalies. The CAD method relies on highly correlated series; thus, one series behavior can be predicted from another series behavior. Increasing the number of inserted anomalies may reduce the correlation which reduces the recall, so that this method is not very effective when inserting a higher number of anomalies.

B. Detecting Anomalies in QSM Dataset

Using the weekly QSM sectors' data, our experiments shown in Fig.3. illustrate the results of the CAD method, CAD combined with our SMA method and the K-mean clustering. Using CAD with SMA, the best recall is achieved which means most anomalies were retrieved. Compared to S&P sectors, the F4 measure in QSM sectors is lower using all algorithms because the precision achieved by these algorithms using QSM data is lower than that of S&P.

The market size in QSM is smaller than in S&P, this may explain why the precision in QSM is lower. In addition to this, QSM companies are younger thus they may behave differently than older and mature companies and they are less correlated than companies in S&P.

VI. CONCLUSION

Valuable insights of strengths and weaknesses of the methods are conducted in our experiments. The analysis of our results suggests when our method can be used and what areas can benefit from further improvement. Although manipulations are well detected using this method, many normal behaviors are reported falsely because the precision of this method is not good enough. A further work to improve precision is recommended to improve the detection system.

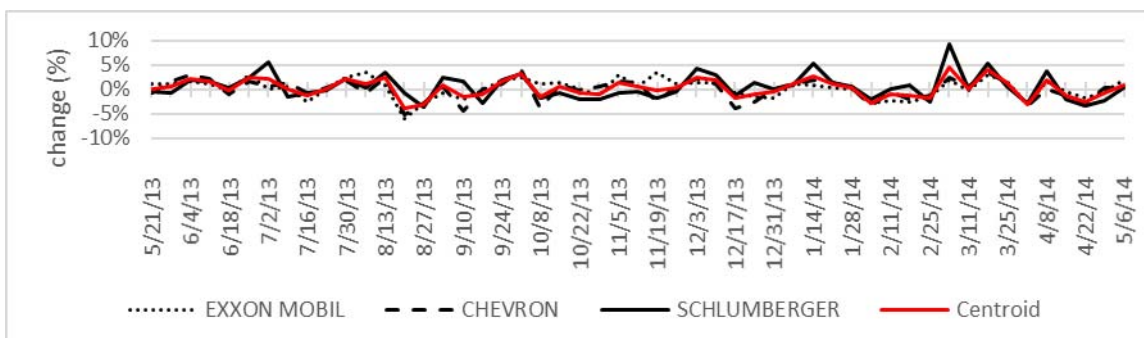


Fig. 2. Centroid Time Series of Energy Sector Stocks in S&P 500

TABLE IV. RESULTS OF CAD, CAD & SMA AND K-MEANS CLUSTERING IN A WEEKLY S&P DATA

S&P 500					
Dataset	Algorithm	Recall (%)	Precision (%)	F2 (%)	F4 (%)
Consumer Staples	CAD	34.7	0.33	1.59	4.86
	CAD & SMA	95.05	4.7	19.62	44.61
	K-means	40.01	3.8	13.77	25.64
Consumer Discretionary	CAD	34.15	0.33	1.6	4.88
	CAD & SMA	88.03	1.5	7.02	20.04
	K-means	40.7	3.6	13.30	25.34
Energy	CAD	34.49	0.33	1.58	4.83
	CAD & SMA	86.7	7.4	27.58	53.18
	K-means	40.2	1.5	6.53	15.97
Financial	CAD	35.47	0.34	1.65	5.05
	CAD & SMA	97.23	2.9	12.95	33.37
	K-means	40.74	3.62	13.35	25.41
IT	CAD	33.69	0.34	1.63	4.98
	CAD & SMA	96.41	6.92	26.88	54.76
	K-means	18.32	0.83	3.51	8.18

TABLE V. THE APPLICATION OF CAD-SMA USING VARIED NUMBERS OF ANOMALIES ON CONSUMER DISCRETIONARY WEEKLY

Anomaly %	Recall	Precision	F2	F4
0.10%	95.05	4.7	19.62	44.61
5%	68.66	73.27	69.54	44.61
10%	54.64	77.86	58.11	68.92
25%	42.44	84.5	47.13	55.62

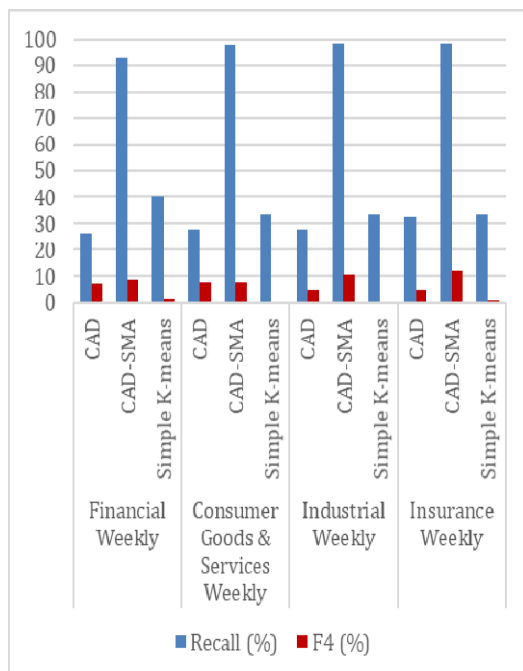


Fig. 3. Results of CAD, CAD - SMA and K-means clustering in a Weekly S&P Data

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